



BANTAY BARYA

Complete User Manual & Technical Reference Guide

VERSION 2.9.0 • DEEP TEAL / FINTECH EDITION

■ Comprehensive System Documentation

- **Architecture & Local-First Philosophy:** 100% client-side privacy, zero cloud storage requirements, and encrypted offline operation.
- **Multi-Wallet & Multi-Currency Engine:** Liquid cash, digital banks, PDC checking accounts, live FX conversions, and zero-expense inter-wallet transfers.
- **Financial Runway & Buffer Tracking:** Rule 4 daily outflow analysis with real-time health indicator thresholds.
- **Liabilities & Snowball Payoff Simulator:** Effective Interest Rate (EIR % p.a.) calculator and 13th-month bonus lump-sum debt acceleration.
- **Accounting Balance Sheet:** Real-time verification of $\text{Total Assets} = \text{Total Liabilities} + \text{Real Net Worth}$
- **Multi-Profile Save Vault:** Snapshot management, JSON exports, and portable `.barya` archives.

Designed and product-directed by **Jerome Gotangco** (<https://github.com/jgotangco>).
Developed with **Google Antigravity / Gemini**.

Visual Direction:
unslop.site



Table of Contents

Chapter	Topic	Key Functional Area
1	Introduction & Core Philosophy	Local-First, Privacy, Zero Cloud Dependency
2	Visual Direction & Design System	Deep Teal Palette, Saturated Fintech Tokens
3	Quick Start & Onboarding	Welcome Modal, Saved Ledger Detection & Safeguards
4	Multi-Wallet Architecture & Currencies	Asset Classification, Transfers, FX Conversions
5	Transaction Logging & Ledger Operations	Continuous Balance, Category Datalists, Sorting
6	Balance Reconciliation (💛 Two Hands)	Bank Statement Matching, Adjustment Auditing
7	Spending Buffer Habit (Rule 4)	Daily Outflow Rate, Financial Runway Days
8	Debt Payoff & Snowball/Avalanche	EIR % Calculations, 13th-Month Pay Acceleration
9	Recurring Bills & Due Dates Calendar	Utilities Schedule, Due Warnings, Auto-Posting
10	Reports & Accounting Balance Sheet	Expense Doughnuts, Statement of Financial Position
11	Save Vault & Multi-Profile Ledger (.barya)	Multiple Budgets, Instant Snapshots, Encrypted Files
12	Security, PIN Lock & Privacy	7-Digit Lockout Keypad, Offline Encryption
13	PWA & Browser Extension Integration	Desktop App Installation, E-Commerce Shopping Logger
14	Philippine Financial Context & Strategies	Emergency Funds, Sandwich Generation, 50/30/20 Rule
15	Keyboard Shortcuts, FAQ & Support	Hotkeys, Data Portability, Troubleshooting

1. Introduction & Core Philosophy

Bantay Barya (Filipino for "*Coin Sentinel*" or "*Guard of Every Cent*") is an offline-capable, local-first single-page web application engineered to grant individuals and families absolute clarity, control, and resilience over their finances.

**100% Client-Side Privacy**

Your financial transactions, wallet balances, loans, and personal notes remain exclusively inside your browser's local storage. No remote servers, no third-party tracking, and zero mandatory user signups.

**Continuous Real-Time Math**

Every transaction instantly recalculates running balances, spending buffer days, net operating surplus, and formal balance sheet equity across all currencies simultaneously.

2. Visual Direction & Design System

Bantay Barya is styled with the **Deep Teal / Fintech** visual direction (credited to [unslop.site](#)), utilizing saturated contrast for maximum readability in high-density data environments:

Token	Hex Value	Usage Context
--bg-app	#07343e	Deep Saturated Teal Canvas background
--bg-surface	#052a33	Dark Petrol primary card containers and ledger rows
--accent-primary	#7be3a8	Electric Spring Mint for CTA buttons, inflows, and active tabs
--debit-color	#ff7b92	Rose Coral for expense debits, liabilities, and warnings
--border-color	rgba(123, 227, 168, 0.2)	Luminous translucent glass borders

3. Quick Start & Onboarding

When launching Bantay Barya, the intelligent **Welcome Dialog** detects whether active financial records exist in your browser storage:

- **Active Saved Ledger Hero Card:** Displays your total converted net balance, wallet count, transaction count, and relative last saved timestamp (*"Saved 2m ago"*). Click  **Load Latest Saved Ledger** to immediately resume your session.
- **Start Fresh:** Clears existing demo records and initializes your primary personal wallet.
- **Try Sample Data:** Loads pre-configured Philippine banks, e-wallets, foreign currencies, recurring utility bills, and loans.
- **Restore Backup:** Uploads any previous `.json` or `.barya` backup file.

 **Active Ledger Overwrite Safeguard:** Destructive operations (Start Fresh, Try Sample Data, Restore Backup) prompt a safeguard confirmation modal summarizing the active ledger that would be replaced, and provide a  **Download Backup First (.barya)** button to guarantee you never lose your records.

4. Multi-Wallet Architecture & Multi-Currency Engine

Rather than dumping all funds into a single balance, Bantay Barya enforces a bucket-based multi-wallet architecture:

Category	Icons	Representative Accounts
Checking & Current		BPI, BDO, Metrobank, UnionBank (with PDC check tracking)
High-Yield Savings		Digital banks (Maya 6%, SeaBank, GoTyme, Tonik, CIMB)
E-Wallets		GCash, Maya Everyday, GrabPay
Cash on Hand		Physical cash in wallet, envelope emergency cash
Fixed Income & Bonds		Pag-IBIG MP2, RTB-30 Treasury Bonds, Time Deposits
Equities & Crypto		PSE stock portfolios, Global ETFs, Cold storage crypto

⇒ **Zero-Expense Inter-Wallet Transfers:** Transferring money between your own wallets (e.g. BPI to GCash) is recorded as linked `transfer_out` and `transfer_in` transactions. These transfers update wallet balances without inflating your monthly expense reports or spending buffer!

5. Continuous Transaction Logging & Ledger Operations

Transactions can be logged in under two seconds through the streamlined entry form:

- **Type Selector:** Toggle between **Debit (Expense)** and **Credit (Income)**.
- **Wallet Selection:** Select the originating or destination account.
- **Quick Date Chips:** Tap **Today** or **Yesterday** for instant date setting.
- **Category Datalist:** Select from categorized presets (Food, Utilities, Transportation, Rent) or type a custom category.
- **Resulting Balance Preview:** Displays what the target wallet's balance will become before you hit save.
- **Ledger Tools:** Filter by wallet, entry type, or timeframe (This Month, Last Month, This Year), with multi-column ascending/descending sorting.

6. Balance Reconciliation Workflow (💛 Two Hands Shaking)

Reconciliation guarantees that your digital ledger matches your actual physical cash and official bank statements down to the exact centavo:

1. Click the 💛 **Reconcile** button on the Wallets Bar or next to any wallet card.
2. Enter your **Actual Statement Balance** directly from your bank app or counted cash.
3. Bantay Barya compares the computed ledger balance against your statement. If a discrepancy exists, it automatically records a categorized **Balance Reconciliation** entry to balance the accounts.

7. Spending Buffer Habit (Rule 4 Financial Runway)

The **Spending Buffer** metric (derived from the *Age of Money* rule) calculates how many days your family can survive on existing liquid cash reserves:

$$\text{Spending Buffer (Days)} = \frac{\text{Total Liquid Converted Assets}}{\text{Average Daily Outflows (Last 30 Days)}}$$

Indicator	Buffer Duration	Financial Status & Strategy
● Healthy	≥ 30 Days	Excellent cash runway. You are spending money earned over a month ago, breaking the paycheck-to-paycheck trap!
● Moderate	15 – 29 Days	Adequate buffer. Continue funneling income into high-yield emergency funds.
● Critical	< 15 Days	Vulnerable position. Immediately trim discretionary debits and postpone non-essential purchases.

8. Debt Payoff & Snowball/Avalanche Engine

Access  **Debts & Payoff** to track all loans (Credit Cards, Auto Loans, Pag-IBIG Mortgages, SSS/GSIS Salary Loans) with precise Effective Interest Rate (EIR % p.a.) mathematical simulation:

-  **Debt Snowball Strategy:** Focuses extra payments on the smallest balance first for rapid psychological momentum.
-  **Debt Avalanche Strategy:** Focuses extra payments on the highest interest rate (EIR %) loan first to mathematically minimize total interest paid.
- **13th-Month Pay & Lump Sum Simulation:** Adjust the extra payment slider or enter year-end bonus lump sums to see your exact accelerated Debt-Free Date!

9. Recurring Bills & Subscriptions Calendar

Manage monthly obligations via  **Bills & Due Dates**:

- Set due dates, auto-pay flags, and reminder warning thresholds (e.g. alert 3 days before due date).
- Header notification badge illuminates with overdue or upcoming bills.
- One-tap bill payment posting directly debits your chosen wallet and records the ledger transaction automatically.

10. Financial Reports & Statement of Financial Position

Open **Reports** to access executive-level financial analytics and formal accounting statements:

Expense Category Breakdown

Interactive Doughnut, Pie, and Bar charts illustrating spending share percentages and transaction volume by category.

Accounting Balance Sheet

A formal Statement of Financial Position verifying that

Total Assets = Total Liabilities + Real Net Worth with full print layout support.

Real Net Worth = Total Assets (Liquid + Investments) - Total Liabilities (Debts + Overdrafts)

11. Save Vault & Multi-Profile Ledger System (`.barya`)

The  **Save Vault** feature allows managing multiple independent ledgers:

- **Multiple Slots:** Create isolated profiles for *Personal*, *Side Business*, *Vacation Budget*, or *What-If Scenarios*.
- **Instant Snapshots:** Duplicate or archive a snapshot before major financial adjustments.
- **Portable `.barya` Files:** Export complete, encrypted ledger snapshots that bundle all wallets, transactions, debts, bills, and settings into a single portable file.

12. Security, PIN Lock & Privacy

- **7-Digit PIN Lockout:** Enable in Settings (⚙️). Intercepts app startup with an on-screen keypad and physical numeric key listener.
- **No Server Transmission:** Encrypted purely in local browser storage.

13. PWA Installation & Browser Extension Integration

- **Desktop (Chrome/Edge):** Click the Install icon in the browser address bar to install Bantay Barya as a standalone windowed desktop app.
- **Mobile (iOS/Android):** Open in Safari or Chrome → Tap **Share / Menu** → **Add to Home Screen**.
- **Browser Extension:** Located in `/extension` folder. Log online shopping purchases (Shopee, Lazada, Amazon) with 1 click!

14. Philippine Financial Context & Best Practices

- 1. **3-to-6 Months Emergency Fund:** Store liquid cash in high-yield digital banks (Maya 6%, SeaBank, GoTyme) earning 4%-6% p.a. rather than traditional low-interest savings accounts (0.0625% p.a.).
- 2. **The 13th-Month Pay 50/30/20 Strategy:** Allocate 50% to high-rate debt reduction, 30% to savings/MP2, and 20% to holiday celebrations.
- 3. **Truth in Lending Act (R.A. 3765) Awareness:** Beware of "1% monthly add-on" advertisements; always request the true Effective Interest Rate (EIR %) disclosure.
- 4. **Full Statement Balance Payments:** Always pay 100% of credit card statements monthly to avoid compounding 3.0%/month (36% EIR) finance charges.

15. Keyboard Shortcuts, FAQ & Troubleshooting

Shortcut	Action
Ctrl + Enter	Save & Record Transaction immediately from form
Escape	Close any active modal dialog or popup overlay
Tab / Shift + Tab	Navigate through transaction form fields and date chips

Data Backup Recommendation: Periodically export your `.barya` backup archive from Save Vault and store it in your synced Google Drive, USB drive, or offline folder for permanent safety!